2.3.2 Parallel transport $\prod_c^\nabla$ along a smooth curve c

Since the manifold is not embedded[1] in a Euclidean space, we cannot add a vector $v \in T_p$ to a vector $v' \in T_{p'}$ as the tangent vector spaces are unrelated to each others without a connection.[2] Thus a *connection* ∇ defines how to associate vectors between infinitesimally close tangent planes T_p and $T_{p+\mathrm{d}p}$. Then the connection allows us to smoothly *transport* a vector $v \in T_p$ by sliding it (with infinitesimal moves) along a smooth curve $c(t)$ (with $c(0) = p$ and $c(1) = q$), so that the vector $v_p \in T_p$ "corresponds" to a vector $v_q \in T_q$: This is called the *parallel transport*. This mathematical prescription is necessary in order to study dynamics on manifolds (e.g., study the motion of a particle[3] on the manifold). We can express the parallel transport along the smooth curve c as:

$$\forall v \in T_p, \forall t \in [0,1], \quad v_{c(t)} = \prod_{c(0)\to c(t)}^{\nabla} v \in T_{c(t)} \tag{16}$$

The parallel transport is schematically illustrated in Figure 3.

2.3.3 ∇-geodesics γ_∇: Autoparallel curves

A connection ∇ allows one to define ∇-*geodesics* as autoparallel curves, that are curves γ such that we have:

$$\nabla_{\dot\gamma}\dot\gamma = 0. \tag{17}$$

That is, the *velocity vector* $\dot\gamma$ is moving along the curve parallel to itself (and all tangent vectors on the geodesics are mutually parallel): In other words, ∇-geodesics generalize the notion of "straight Euclidean" lines. In local coordinates $(\mathcal{U}, x)$, $\gamma(t) = (\gamma^k(t))_k$, the autoparallelism amounts to solve the following second-order Ordinary Differential Equations (ODEs):

$$\ddot\gamma(t) + \Gamma^k_{ij}\dot\gamma(t)\dot\gamma(t) = 0, \quad \gamma^l(t) = x^l \circ \gamma(t), \tag{18}$$

where Γ^k_{ij} are the *Christoffel symbols of the second kind*, with:

$$\Gamma^k_{ij} \overset{\Sigma}{=} \Gamma_{ij,l} g^{lk}, \quad \Gamma_{ij,k} \overset{\Sigma}{=} g_{lk}\Gamma^l_{ij}, \tag{19}$$

where $\Gamma_{ij,l}$ the *Christoffel symbols of the first kind*. Geodesics are 1D autoparallel submanifolds and ∇-hyperplanes are defined similarly as autoparallel submanifolds of dimension $D-1$. We may specify in subscript the connection that yields the geodesic γ: γ_∇.

The geodesic equation $\nabla_{\dot\gamma(t)}\dot\gamma(t) = 0$ may be either solved as an *Initial Value Problem* (IVP) or as a *Boundary Value Problem* (BVP):

- Initial Value Problem (IVP): fix the conditions $\gamma(0) = p$ and $\dot\gamma(0) = v$ for some vector $v \in T_p$.
- Boundary Value Problem (BVP): fix the geodesic extremities $\gamma(0) = p$ and $\gamma(1) = q$.

2.3.4 Curvature and torsion of a manifold

An affine connection ∇ defines a 4D[4] *curvature tensor* R (expressed using components R^i_{jkl} of a $(1,3)$-tensor). The coordinate-free equation of the curvature tensor is given by:

$$R(X,Y)Z := \nabla_X\nabla_Y X - \nabla_Y\nabla_X Z - \nabla_{[X,Y]}Z, \tag{20}$$

[1] Whitney embedding theorem states that any D-dimensional Riemannian manifold can be embedded into $\mathbb{R}^{2D}$.

[2] When embedded, we can implicitly use the ambient Euclidean connection ${}^{\mathrm{Euc}}\nabla$, see [2].

[3] Elie Cartan introduced the notion of affine connections [27, 3] in the 1920's motivated by the *principle of inertia* in mechanics: A point particle, without any force acting on it, shall move along a straight line with constant velocity.

[4] It follows from symmetry constraints that the number of independent components of the Riemann tensor is $\frac{D^2(D^2-1)}{12}$ in D dimensions.

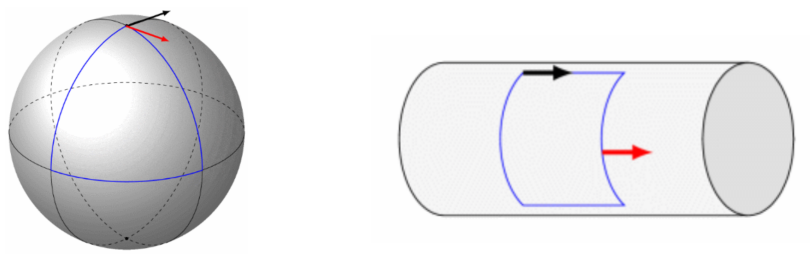

Figure 4: Parallel transport with respect to the metric connection: Curvature effect can be visualized as the angle defect along the parallel transport on smooth (infinitesimal) loops. For a sphere manifold, a vector parallel-transported along a loop does not coincide with itself, while it always conside with itself for a (flat) manifold. Drawings are courtesy of © CNRS, `http://images.math.cnrs.fr/Visualiser-la-courbure.html`

where $[X,Y](f) = X(Y(f)) - Y(X(f))$ $(\forall f \in \mathfrak{F}(M))$ is the *Lie bracket* of vector fields. When the connection is the metric Levi-Civita, the curvature is called *Riemann-Christoffel curvature tensor*. In a local coordinate system, we have:

$$R(\partial_j, \partial_k)\partial_i \overset{\Sigma}{=} R^l_{jki}\partial_l. \tag{21}$$

Informally speaking, the curvature tensor as defined in Eq. 20 quantifies the amount of non-commutativity of the covariant derivative.

A manifold M equipped with a connection ∇ is said *flat* (meaning ∇-flat) when $R = 0$. This holds in particular when finding a *particular*[5] coordinate system x of a chart $(\mathcal{U}, x)$ such that $\Gamma^k_{ij} = 0$, i.e., when all connection coefficients vanish.

A manifold is *torsion-free* when the connection is *symmetric*. A symmetric connection satisfies the following coordinate-free equation:

$$\nabla_X Y - \nabla_Y X = [X, Y]. \tag{22}$$

Using local chart coordinates, this amounts to check that $\Gamma^k_{ij} = \Gamma^k_{ji}$. The torsion tensor is a $(1,2)$-tensor defined by:

$$T(X,Y) := \nabla_X Y - \nabla_Y X - [X,Y]. \tag{23}$$

For a torsion-free connection, we have the first Bianchi identity:

$$R(X,Y)Z + R(Z,X)Y + R(Y,Z)X = 0, \tag{24}$$

and the second Bianchi identity:

$$(\nabla_V R)(X,Y)Z + (\nabla_X R)(Y,V)Z + (\nabla_Y R)(V,X)Z = 0. \tag{25}$$

In general, the parallel transport is *path-dependent*. The *angle defect* of a vector transported on an *infinitesimal closed loop* (a smooth curve with coinciding extremities) is related to the curvature. However for a *flat connection*, the parallel transport does not depend on the path, and yields *absolute parallelism geometry* [133]. Figure 4 illustrates the parallel transport along a curve for a curved manifold (the sphere manifold) and a flat manifold (the cylinder manifold[6]).

An affine connection is a torsion-free linear connection. Figure 5 summarizes the various concepts of differential geometry induced by an affine connection ∇ and a metric tensor g.

[5]For example, the Christoffel symbols vanish in a rectangular coordinate system of a plane but not in the polar coordinate system of it.

[6]The Gaussian curvature at of point of a manifold is the product of the minimal and maximal sectional curvatures: $\kappa_G := \kappa_{\min}\kappa_{\max}$. For a cylinder, since $\kappa_{\min} = 0$, it follows that the Gaussian curvature of a cylinder is 0. Gauss's Theorema Egregium (meaning "remarkable theorem") proved that the Gaussian curvature is intrinsic and does not depend on how the surface is embedded into the ambient Euclidean space.

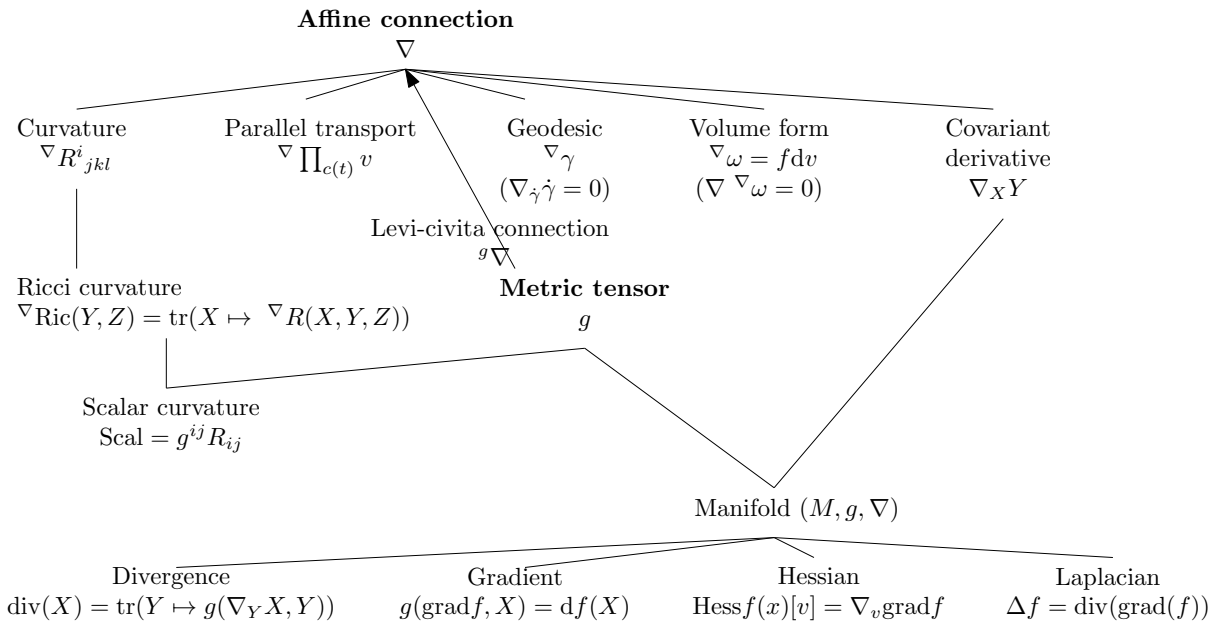

Figure 5: Differential-geometric concepts associated to an affine connection ∇ and a metric tensor g.

Curvature is a fundamental concept inherent to geometry [22]: There are several notions of curvatures: scalar curvature, sectional curvature, Gaussian curvature of surfaces to Riemannian-Christoffel 4-tensor, Ricci symmetric 2-tensor, synthetic Ricci curvature in Alexandrov geometry, etc.

2.4 The fundamental theorem of Riemannian geometry: The Levi-Civita metric connection

By definition, an affine connection ∇ is said *metric compatible* with g when it satisfies for any triple (X, Y, Z) of vector fields the following equation:

$$X\langle Y, Z\rangle = \langle \nabla_X Y, Z\rangle + \langle Y, \nabla_X Z\rangle, \tag{26}$$

which can be written equivalently as:

$$Xg(Y, Z) = g(\nabla_X Y, Z) + g(Y, \nabla_X Z) \tag{27}$$

Using local coordinates and natural basis $\{\partial_i\}$ for vector fields, the metric-compatibility property amounts to check that we have:

$$\partial_k g_{ij} = \langle \nabla_{\partial_k}\partial_i, \partial_j\rangle + \langle \partial_i, \nabla_{\partial_k}\partial_j\rangle \tag{28}$$

A property of using a metric-compatible connection is that the parallel transport $\prod^{\nabla}$ of vectors preserve the metric:

$$\langle u, v\rangle_{c(0)} = \left\langle \prod_{c(0)\to c(t)}^{\nabla} u, \prod_{c(0)\to c(t)}^{\nabla} v \right\rangle_{c(t)} \quad \forall t. \tag{29}$$

That is, the parallel transport preserves angles (and orthogonality) and lengths of vectors in tangent planes when transported along a smooth curve.

The fundamental theorem of Riemannian geometry states the existence of a unique torsion-free metric compatible connection: